

LEVENT ÇELİKSAN

Principal AI Architect

Technical Founder of Sealify | Agentic AI · Computer Vision · AI Systems

Istanbul, Türkiye | Open to remote work | Open to relocating to the US (requires visa sponsorship)
leventceliksan@gmail.com | linkedin.com/in/levent-celiksan | github.com/LeventCeliksan | sealify.io

SUMMARY

Principal AI Architect and hands-on technical founder with 19+ years of backend and systems engineering, including leading and mentoring engineering teams. Took Sealify from idea to public launch as its sole technical founder. Focused on applied AI engineering: production systems validated by measurement on real-world web data rather than research prototypes — agentic AI, computer vision, multimedia matching, KYC/identity verification, and content authentication.

As AI tools make writing code easier every day, the real differentiator is deciding which architecture a system should use, how secure it must be, what hardware and server capacity it needs, what it costs per unit, and how sustainably it can run. At Sealify, owned these decisions end to end, from architecture and technical strategy through implementation, validation, infrastructure, and production operations; filed six US provisional patent applications in AI-based content protection, deepfake detection, and multimedia matching.

FOUNDER-ENGINEER STRENGTHS

- **End-to-end ownership:** Took Sealify from idea to production as the sole technical founder, owning product and system architecture, software development, infrastructure, and operations.
- **Cost and unit economics:** Saved roughly \$1 per verification versus paid KYC providers through self-hosted face matching; designed Bitcoin-based timestamping without passing transaction fees to users; combined free and paid search methods with automatic failover; moved heavy inference to user devices, eliminating the need for server-side inference.
- **Security and data privacy:** Built a self-hosted KYC pipeline that processes identity and biometric data on its own infrastructure and a privacy-oriented, on-device architecture that never sends images to a server; handled Linux server security, backups, and monitoring.
- **Hardware, infrastructure, and capacity:** Planned RAM, CPU, and GPU usage, server capacity, and energy cost together; chose between AWS, GCP, Azure, Vultr, and self-hosted infrastructure by workload; built queue-based, multi-instance architectures.
- **AI architecture decisions (build vs. buy):** Decided which tasks run on open-source/local models and which use paid APIs; implemented multi-LLM routing by task type and quota status; split inference between on-device and server-side execution.
- **IP, finance, and investor relations:** Inventor on six US provisional patent applications; held technical and operational responsibility in financial planning and investor meetings.

PROFESSIONAL EXPERIENCE

Technical Founder & Principal AI Architect — Sealify, Inc.

Nov 2023 – Present (incorporated in Delaware in 2025) | Newark, Delaware (Remote)

- Launched Sealify publicly in June 2026 as a multilingual (English, Turkish, Arabic) SaaS platform and onboarded ~100 early users; the platform includes Google sign-in, tiered subscription plans (free to \$200/month) with crypto payments, a user dashboard, and an internal admin panel for user management, credits, email, and scheduled DMCA takedown scans. Demo video: <https://www.youtube.com/watch?v=Ll0T1nJU5c> · Intro video: <https://www.youtube.com/watch?v=Bh7kB-UIJhs>
- Validated Sealify's internet-scale content scanning, video matching, and autonomous social media archiving systems through 3,000+ measured test runs. Scanned 200,000+ candidate images and identified 10,000+ verified matches across 5,000+ domains. A single video test found 1,050+ matches across 80+ domains. Social media archiving tests automatically processed and registered 10,000+ media items and 450+ GB of data.
- Developed a self-hosted identity verification and KYC face-matching pipeline that compares ID or passport photos with selfies using open-source face matching models, reducing reliance on paid providers such as Onfido or Jumio and saving roughly \$1 per verification. Designed a camera-only capture flow (no file uploads) for selfies and ID/passport documents, including MRZ, to reduce spoofing and replay attacks.
- Built a free timestamping system that anchors content hashes to the Bitcoin blockchain using OpenTimestamps. Combined SHA-256 hashing with perceptual-hash fingerprinting so content can be timestamped verifiably, with no transaction fees on the user side. More than 1,600 registrations have been anchored on Bitcoin mainnet in the live system, each independently verifiable on public block explorers.
- Engineered a fine-tuned image and face matching engine using EfficientNet-B7, InsightFace/ArcFace, and color and lighting signals. The system runs on a queue-based, multi-instance architecture for scene and face similarity detection.
- Built a two-tier reverse image/video search engine that runs a free method in parallel with a paid API and fails over automatically when either one hits a rate limit. From a single photo, poster, or video frame, the system finds publicly accessible, detectable copies of content across the web in a median of 2.5 minutes, including versions with changed resolution, removed or altered audio, or heavily altered colors or visual style.

- Developed a hybrid deepfake detection prototype that determines whether images and videos are AI-generated; built a decision engine that combines image-analysis signals (regional blur anomalies, homogeneous texture, noise, color and contrast distribution) with voting across Hugging Face classifiers, plus an asynchronous video analysis service that samples frames from YouTube/TikTok videos.
- Built a system that finds non-consensual intimate and deepfake content by combining web image search (SerpApi/Yandex) with InsightFace/ArcFace face matching, and shipped it as a paid-plan feature for scanning adult content platforms; designed a privacy-focused flow that processes hundreds of images per scan using parallel downloads and Apple Silicon (CoreML) acceleration, displays results blurred, and deletes data automatically.
- Built autonomous worker systems that check Instagram, TikTok, and YouTube channels every 15 minutes, automatically downloading newly uploaded content, extracting its fingerprint, and registering it on the blockchain.
- Built a scraper that continuously monitors pirate movie sites and feeds newly listed titles into the detection pipeline in near real time.
- Built an AI research agent using LangGraph and local LLMs that autonomously investigates, evaluates, and reports on suspected piracy sites end to end; the system generates evidence-based reports automatically.
- Made technical decisions based not only on software performance but also on the company's long-term financial sustainability. Planned infrastructure end to end, from RAM/CPU/GPU usage to server capacity and third-party service costs, creating a scalable, predictable, low-cost production environment.

SELECTED AI PROJECTS

Aesthetic Detector — BEAUTY (Android)

2025 – 2026 | *Personal product: Android app built in Kotlin with a credit-based revenue model; preparing for Google Play launch*

- Fine-tuned a CLIP ViT-B/16-based model on 1,000+ portrait photos to build a deep learning system that classifies signs of cosmetic intervention, then converted the model to ONNX and optimized it to run inference entirely on-device.
- Developed a multi-stage scoring pipeline that combines the overall classification score with EfficientNet-B0-based regional facial analysis; used Google ML Kit for face detection and pose checks and ONNX Runtime for model inference.
- Designed a privacy-oriented architecture that runs heavy AI processing on the user's device so images never need to be sent to a server; used the Node.js, Express, and MySQL backend only for user authentication, the credit system, data synchronization, and coordination.

AI Advertising Agency Automation

2025 – Present

- Built a generative AI advertising agency automation system with 7 agents managed by a central orchestrator, distributing natural-language briefs across copywriting, design, photography, social media, software, video, and assistant roles.
- Built the agent orchestration and backend infrastructure on Python, FastAPI, Pydantic, and SQLite; developed a multi-LLM architecture that routes the Gemini API, local models running on Ollama, and models accessed via OpenRouter by task type, switching to an alternative model on quota limits or errors.
- Developed autonomous quality-control and revision mechanisms that evaluate agent outputs, check brief compliance, reprocess failed results, and rerun only the agent that needs it.
- Designed the system to run most heavy workloads locally, creating a backend architecture that keeps third-party service dependency and ongoing server costs to a minimum.

EARLIER EXPERIENCE

Principal Engineer — BUBU

Apr 2018 – Oct 2023

- Set the technical direction for the agency's client projects; established shared architecture, infrastructure, and security standards, mentored engineers, and provided technical consulting to clients.
- Developed REST APIs with Python (Flask, FastAPI), MySQL, and PostgreSQL, and e-commerce backends covering product catalog, inventory, checkout, orders, payments, and marketplace integrations.
- Owned production infrastructure and IT operations for clients including Kopuzlar Holding; set up Linux server security, Docker deployments, CI/CD, monitoring, and backup systems.
- Built the first LLM integrations (OpenAI ChatGPT API) and RAG prototypes for client projects in 2023.

Lead Software Engineer — Artworks Lab AB

Sep 2015 – Jan 2018

- Led the backend team; managed sprint planning, estimation, code review, mentoring, and hiring.
- Built Django and Django REST Framework backends for e-commerce platforms with payment, shipping, and ERP integrations; scaled them for campaign-period traffic with Elasticsearch and Redis, and deployed Docker services to production on AWS.

Senior Software Engineer — Yellow Pages / DataWorks

Jan 2012 – Jun 2015

- Designed the REST APIs and phone number lookup service behind the large-scale phone directory of Kim Ariyor, a caller-ID app for Android and iOS.
- Built directory search, spam number reporting, and Google Maps API-based location features with PHP/Laravel; scaled the MySQL data layer and ran AWS server operations.

Software Engineer — Jolly Tur

Mar 2008 – Dec 2011

- Built online booking backends with PHP/CodeIgniter and MySQL, XML/SOAP integrations for hotel and flight suppliers, and virtual POS payment systems; owned the company's Linux and Windows Server infrastructure.

Junior Software Engineer — REKLAMARKA

Dec 2006 – Jan 2008

- Built PHP/MySQL backends, admin panels, reporting modules, and SOAP/XML integrations for e-commerce and campaign projects.

PATENT APPLICATIONS

Inventor: Levent Çeliksın

Six US provisional patent applications filed with the USPTO in Nov–Dec 2025. Status: Patent Pending.

- Digital Content Tracking and Protection System — App. No. 63/923,341
- Real-Time Context-Aware Deepfake Detection System with Zero-Permission Floating Trigger and Hybrid Multi-Modal Analysis — App. No. 63/923,358
- Method for Instant Multimodal Novelty Assessment Across Public Networks — App. No. 63/923,342
- Method and System for Autonomous Registration of Subscription-Based Video Content Within a 3-Minute Latency Boundary Using an API-Independent External Protocol — App. No. 63/934,369
- Conditional Routing-Based Multimedia Semantic Equivalence Detection — App. No. 63/946,066
- Artificial Intelligence-Based Virtual Human Marketplace with Telemetry-Driven Dynamic Licensing and Topology-Preserving Automated 3D Character Aging Pipeline — App. No. 63/923,478

Invited to apply to IPHatch North America 2026, Piece Future's patent-commercialization program; Sealify is currently under evaluation.

TECHNICAL SKILLS

- **Generative AI & agent systems:** LLMs, agentic AI, multi-agent systems, LangGraph, LangChain, RAG, tool calling, structured LLM outputs, multi-LLM routing and failover, FAISS, Ollama, Gemini API, OpenRouter, model fine-tuning
- **Machine learning & computer vision:** PyTorch, Hugging Face Transformers, deep learning, model deployment & serving (ONNX, on-device, multi-instance), OpenCV, ONNX Runtime, Google ML Kit, InsightFace/ArcFace, EfficientNet, CLIP, DINOv2, YOLOv8 (NudeNet), face matching, perceptual hashing and content fingerprinting, deepfake and AI-generated content detection (image analysis + classifier ensemble), document forensics (perspective correction, MRZ checksum validation), OCR/Tesseract, Whisper
- **Backend:** Python (FastAPI, Pydantic, Flask, Django/DRF, SQLAlchemy, Alembic, Celery, Gunicorn, pytest, Scrapy), Node.js/Express, SQL, Bash; REST APIs, WebSockets, JWT authentication
- **Mobile:** Android (Kotlin), Room, WorkManager, Google Play Billing, Play Asset Delivery, Google ML Kit, on-device ONNX inference
- **Automation:** Playwright, DrissionPage, Chrome DevTools Protocol, BeautifulSoup4/lxml, browser-use, yt-dlp, SerpApi
- **Data:** MySQL, PostgreSQL, MongoDB, SQLite, Redis, Elasticsearch, RabbitMQ
- **Cloud & DevOps:** AWS, GCP, Azure, Vultr, self-hosted; Linux, Nginx/Apache, Git, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, GitLab CI, Jenkins), Prometheus, Grafana, Sentry, Cloudflare Workers/Tunnel
- **Blockchain & verification:** OpenTimestamps / Bitcoin anchoring, SHA-256 hashing, verifiable certificate generation
- **Systems & operations:** Resource and capacity planning, cost modeling, server security; Linux and Windows Server administration, backups, DNS and email systems

EDUCATION & LANGUAGES

Collège LaSalle, Montréal, Canada: Bachelor's Degree, Computer Science Technology • 2002 – 2006

Languages: Turkish — Native | English — Professional working proficiency

CURRENT FOCUS AREAS

- **AI model training & agents:** Model training and fine-tuning, multi-agent systems, agent orchestration
- **Distributed systems:** Gossip protocols, DHT, CRDT, WebRTC, IPFS
- **Blockchain & cryptography:** Web3 infrastructure, zero-knowledge proofs
- **Correctness & performance:** TLA+ formal verification, eBPF, hardware-aware optimization
- **Virtual reality & virtual humans:** VR (virtual reality), 3D virtual human and character technologies